

Basic Electricity Terms

Conductor – Any material that an electrical current can travel through. Copper is a good conductor.

Insulator – Any material that opposes the flow of current. Ex. Plastic or rubber

Inductance – The field around any energized conductor. It is measured in the ability of a circuit to establish a magnetic field. The number of twists in a copper pair affects inductance on a phone line.

Inductor – Preserves current by storing energy in a magnetic field (ex. Load Coil) Load coils increase inductance

Capacitor – 2 electrical conductors that are separated by an insulator (pair of wires, cables). It stores the electrical charge in a circuit where no current is flowing (i.e. a phone line on hook). It preserves voltage by storing the energy in an electrical field. It blocks DC current and allows AC current to pass since it is continuously reversing voltage.

Capacitance – the ability of a conductor to hold a charge. Things that affect capacitance: the closeness of the pair if wires, the length of the wires and the better insulation between the wires

Resistance – Measured in Ohms, resists the flow of current. Effected by size (gauge), distance, temperature and troubles

Current – Is measured in Amps, is the flow of electrical energy. DC flows in on direction, AC reverses direction. Current flows form (-) to (+)

OHM's Law – Is the relationship between voltage, current and resistance in a circuit. It takes 1 volt to push 1 amp through 1 ohm of resistance.

Series Circuit – An electrical circuit that has 1 continuous path, normally has 2 or more resistors and flows from negative to positive.

Kirchoff's law – in a series circuit the total resistance is the sum of all the individual resistors

$E = I/R$ $E =$ Voltage (volts) $R =$ Resistance(Ohms) $I =$ Current(Amps)

Power (P) – the time rate of doing work, measured in watts (1 watt = 1 volt x 1 Amp)

It requires -48 volts to push 23ma- 50 milliamps through 1500- 1900 ohms of resistance in the local loop to convert mechanical voice to electrical energy

Telephone – mechanical to electrical to mechanical

Current through any wires creates a magnetic field – a coil of wire that increases the magnetic field resists AC voltage and lets DC flow freely

Read over Telephone operation in study guide